

EDGE AI COMPUTER · NVIDIA JETSON ORIN NX

SEALED SINTRONES IBOX-602P-IP66 EDGE AI COMPUTER

An IP66, fanless Jetson Orin NX box that starts at -40 °C, takes two GMSL-2 cameras on coax, and powers them itself. Every connector is M-coded – nothing on this machine shakes loose.



■ WHY THIS COMPUTER

- **NVIDIA Jetson Orin NX, 16 GB or 8 GB** – 1024-core Ampere GPU with 32 Tensor Cores, 8- or 6-core Arm Cortex-A78AE at 2.0 GHz, running JetPack 6.2
- **2 × GMSL-2 camera inputs over FAKRA** – synchronized coax capture from the qualified oToBrite, e-con Systems and StereoLabs modules, up to 8 MP per channel
- **IP66 sealed aluminium enclosure, fanless** – dust-tight and rated against powerful water jets, with no moving parts to service
- **Built-in low-temperature heater** – cold start from -40 °C and operation to 65 °C with 0.6 m/s airflow
- **M-coded connectors throughout** – M12 X-code Ethernet, M12 K-code power, M8 for CAN FD, COM and DIO; no RJ-45 to vibrate out of its latch
- **2 × PoE at 15.4 W per port** (PSE Class 3, 32 W budget) – the cameras draw from the same box that processes them
- **9–60 V DC input** with OCP, surge and reverse-polarity protection, ignition sensing, and an optional BBU for 5–10 minutes of orderly shutdown
- **EN 50155, EN 45545-2 (R25), E-mark, CE, FCC Class A and UKCA**, with MIL-STD-810G shock and vibration qualification

IP66

INGRESS PROTECTION

-40 °C

COLD START

2 × GMSL-2

CAMERA INPUTS

9–60 V

DC INPUT RANGE

■ APPLICATION SCENARIOS



AMR & AGV PLATFORMS

On-vehicle inference for autonomous chassis



RAIL & ROLLING STOCK

EN 50155 / EN 45545-2 onboard compute



GMSL-2 VISION

Two synchronized cameras over coax



OFF-HIGHWAY VEHICLES

Sealed compute for outdoor and mobile duty

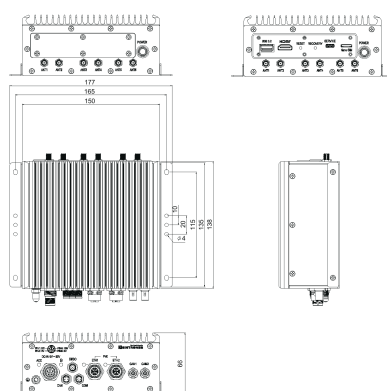
SINTRONES IBOX-602P-IP66 — SPECIFICATIONS



PARAMETER	TECHNICAL DATA
System on module	NVIDIA® Jetson Orin™ NX 16 GB or 8 GB
GPU / accelerators	1024-core NVIDIA® Ampere GPU with 32 Tensor Cores; 2 × NVDLA v2 (16 GB) / 1 × NVDLA v2 (8 GB); 1 × PVA v2
CPU	8-core Arm® Cortex®-A78AE (16 GB SoM) 6-core Arm® Cortex®-A78AE (8 GB SoM) · 2.0 GHz
Memory	16 GB or 8 GB 128-bit LPDDR5 (on SoM)
Storage	1 × M.2 2280 Key M NVMe SSD (pre-installed with system BSP)
Camera	2 × GMSL-2 via FAKRA
Ethernet	1 × 2.5GbE (Intel® I226-IT), 1 × GbE (SoM), both via M12 X-code
PoE	15.4 W per port (PSE Class 3); 32 W total power budget
CAN	1 × CAN FD via M8
Serial port	1 × RS-232/422/485 via M8
DIO	3 × DI, 4 × DO (12 V DC / 100 mA) via M8
USB	1 × USB 3.2 (with cover); 1 × USB Type-C for system recovery (device only)
Video / audio	1 × HDMI® (with cover); 1 × HD audio from the HDMI®
Internal expansion	1 × M.2 3042/3052 Key B (WWAN, dual nano-SIM) 1 × M.2 2230 Key E (Wi-Fi / BT)

PARAMETER	TECHNICAL DATA
Power input	DC 9–60 V (nominal 12 / 24 / 48 V) via M12 K-code
Power protection / mgmt.	OCP, surge and reverse polarity protection; smart power management; high-capacity coin cell for RTC
Battery backup (BBU)	Optional kit, 5–10 min emergency backup (patent M447854)
Security / watchdog	Platform Security Controller (PSC), Security Engine (SE); automatic reset
Operating system	NVIDIA® JetPack 6.2 or above (Jetson Linux + dev tools)
Operating temperature	Startup –40 °C to 65 °C; operating –35 °C to 65 °C (2 × GMSL-2); –35 °C to 60 °C under stress (2 × GMSL-2 + 2 × PoE), 0.6 m/s airflow
Storage temp. / humidity	–40 °C to 80 °C; 10% to 90% RH, non-condensing
Vibration / shock	IEC 60068-2-64 random 2.5 G @ 5–500 Hz, 1 h/axis; MIL-STD-810G 514.6 Cat. 4 and 516.6 Proc. I, 15 G (11 ms)
Certification	CE, FCC Class A, UKCA, E-mark, EN 50155, EN 45545-2 (R25)
Ingress protection	IP66
Construction	Aluminium alloy; wall mounting; 6 × SMA antenna mounting holes
Weight	1.78 kg (3.92 lb)
Dimensions (L × W × H)	150 × 138 × 66 mm (6.0 × 5.5 × 2.6 in)

OUTLINE AND MOUNTING



Unit : mm

UNIT: MM · DRAWINGS NOT TO SCALE

Ordering. IBOX-602P-IP66-ONX16 — Jetson Orin NX 16 GB / 2 × PoE / 2 × GMSL-2. IBOX-602P-IP66-ONX8 — Jetson Orin NX 8 GB / 2 × PoE / 2 × GMSL-2. An M.2 2280 Key M NVMe SSD is a mandatory accessory. Optional: Wi-Fi (M.2 2230 Key A-E), WWAN modem (M.2 3042/3052), GNSS (M.2 2242 Key B), BAT-2300v2 battery kit, and a 100–240 V / 24 V 160 W power adapter — each sold separately.

Notes. Operating temperature figures assume the built-in low-temperature heater is enabled and 0.6 m/s airflow, without stress testing except where stated. BBU backup time varies with actual system power consumption; the battery must be charged before use as a backup source. Qualified GMSL-2 camera support may vary by BSP version — confirm your camera against the current supported list before design-in.

Installation. Mount on a flat surface with the fin stack unobstructed and the specified ambient airflow available; these are fanless chassis and shed heat through the enclosure. Keep the connector face clear for service access. Confirm the DC supply stays within the rated input window under vehicle crank and load-dump conditions before commissioning.

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